MATH 111 Calculus 1

Maximum and minimum values

Week 6 · Lecture 15 · §3.1

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

STAND UP

"If $f'(c) = 0$, then f has a local maximum or a local minimum at c."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

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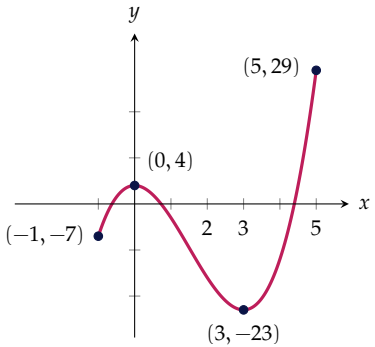
Verdict

FALSE. $f(x) = x^3$: $f'(x) = 3x^2$, so $f'(0) = 0$. But $x^3 < 0$ for $x < 0$ and $x^3 > 0$ for $x > 0$: near 0 the function takes values below *and* above $f(0) = 0$ — no maximum and no minimum at 0. The true statement goes the other way, once $f'(c)$ exists (Fermat's Theorem, frame 8): a local maximum or minimum at c, where $f'(c)$ exists, forces $f'(c) = 0$.

EXAMPLE Our function: $f(x) = 2x^3 - 9x^2 + 4$ on the domain $D = [-1, 5]$ (the picture).

Definition Let c be a number in the domain D of a function f. Then $f(c)$ is the

- absolute maximum value of f on D if $f(c) \geq f(x)$ for all x in D;
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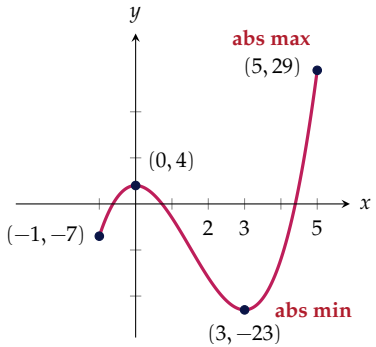
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Absolute extreme values

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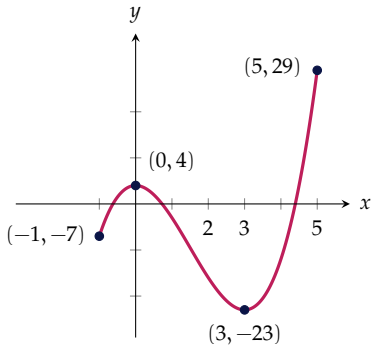
Here: the absolute maximum is $f(5) = 29$; the absolute minimum is $f(3) = -23$.

Also called *global*; together, the *extreme values* of f.

Local extreme values

Definition The number $f(c)$ is a
- local maximum value of f if $f(c) \geq f(x)$ when x is near c;
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"Near c" means: on some open interval containing c.



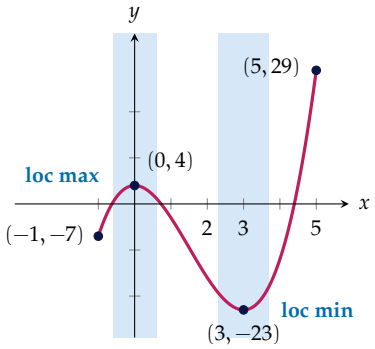
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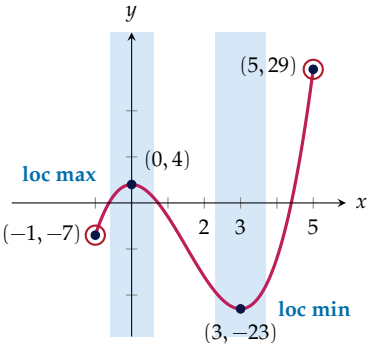
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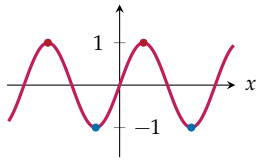
$f(5) = 29$ is absolute but not local: no open interval around the endpoint 5 lies in D. At the endpoint -1: neither.



shaded: open intervals around 0 and around 3; ringed: the endpoints

Not every function has extreme values

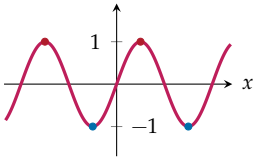
EXAMPLES Three functions on the whole real line.



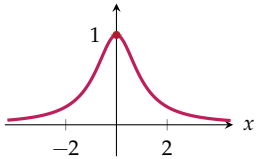
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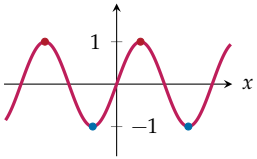
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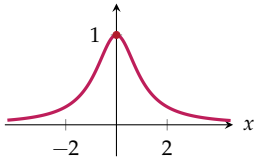
$y = \dfrac{1}{1+x^2}$: maximum 1 at 0, since $1 + x^2 \geq 1$. No minimum: every value is > 0, and the values come as close to 0 as we like.

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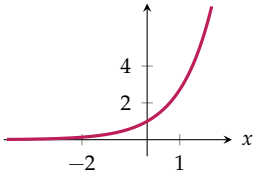
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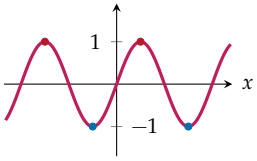
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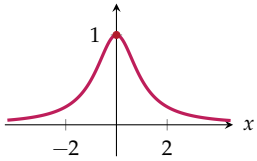
$y = e^x$: increasing, so no maximum and no minimum — not even local ones. The values come as close to 0 as we like, but never reach it.

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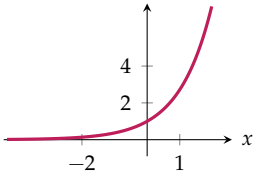
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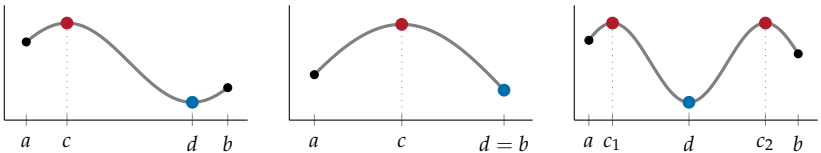
When is a maximum, or a minimum, guaranteed? $\longrightarrow$ the next frame.

The Extreme Value Theorem

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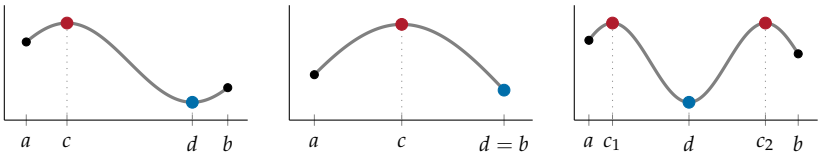
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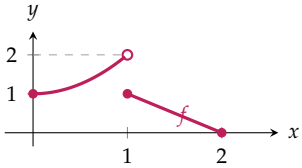
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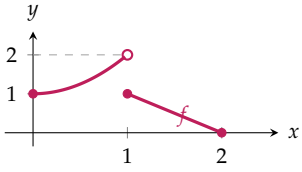
We take this theorem **without proof**: the proof needs a deep property of the real numbers, their *completeness* — the number line has no gaps. It says that the extreme values *exist*, not *where* they are.

Without its hypotheses, no promise

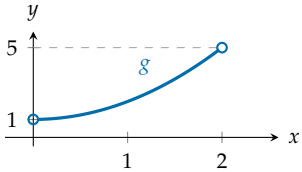


$f(x) = \begin{cases} 1 + x^2, & 0 \le x < 1 \\ 2 - x, & 1 \le x \le 2 \end{cases}$ on the closed interval $[0, 2]$, not continuous at 1. Its values fill $[0, 2)$: as close to 2 as we like, never 2 — no maximum. (A minimum: $f(2) = 0$.)

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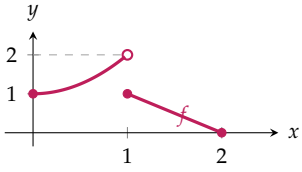


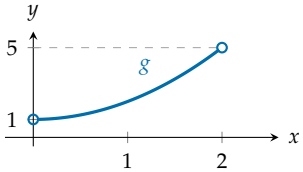
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$g(x) = 1 + x^2$ on the open interval $(0, 2)$: continuous, but the interval is not closed. Its values fill $(1, 5)$: no maximum and no minimum — they would be at the endpoints, which are missing.

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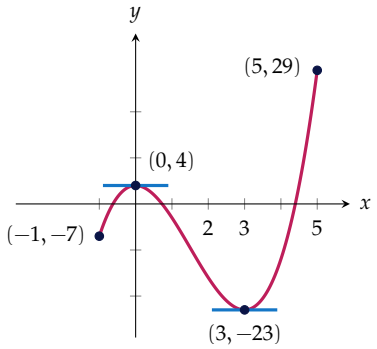
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No contradiction: each time one hypothesis fails. And $1 + x^2$ on the closed interval $[0, 2]$ has both: minimum 1 at 0, maximum 5 at 2.

Fermat's Theorem

The function of frames 3–4 at its local maximum $(0, 4)$ and its local minimum $(3, -23)$: the tangent lines are horizontal — slope 0.

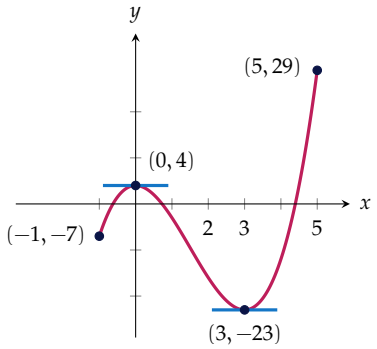


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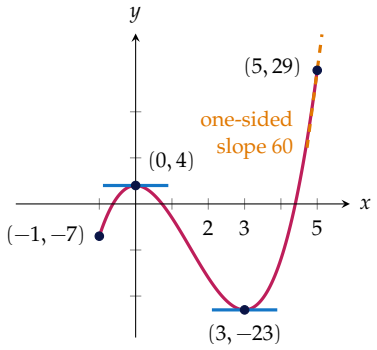
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Check: $f(x) = 2x^3 - 9x^2 + 4$, $f'(x) = 6x^2 - 18x = 6x(x - 3)$: zero exactly at 0 and 3. At the endpoint 5, where the *absolute* maximum is, the one-sided slope is $6 \cdot 5 \cdot 2 = 60$, not 0: an endpoint is never a local extremum, and the theorem says nothing about it.

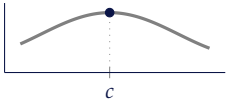


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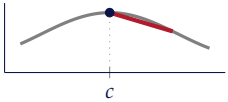
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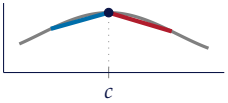


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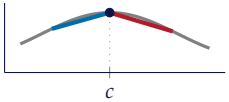
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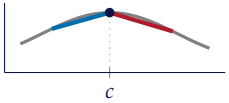
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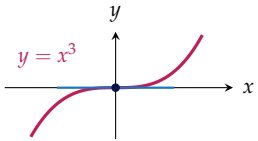
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So $f'(c) \leq 0$ and $f'(c) \geq 0$: $f'(c) = 0$. A local minimum: every inequality reversed. ■

Used: a limit that exists equals each one-sided limit; limits keep $\leq$.

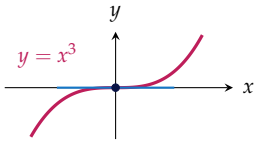
Two cautions

Caution. $f'(c) = 0$ does not force an extreme value. $f(x) = x^3$: $f'(0) = 0$, but $x^3 < 0$ for $x < 0$ and $x^3 > 0$ for $x > 0$ — no maximum, no minimum at 0: the curve crosses its horizontal tangent (the warm-up).

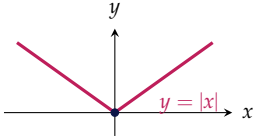


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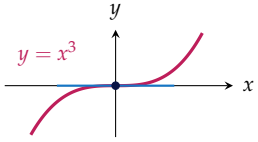


Caution. An extreme value where f' does not exist. $f(x) = |x|$: $f(0) = 0$ is the (local and absolute) minimum, since $|x| \geq 0$; but $f'(0)$ does not exist: $\lim_{h \to 0^+} \frac{|h|}{h} = 1$, $\lim_{h \to 0^-} \frac{|h|}{h} = -1$.

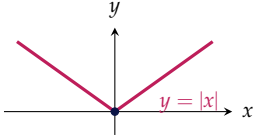


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The converse of Fermat's Theorem is false, and the theorem says nothing where f' does not exist. It does tell us where to start looking: where $f'(c) = 0$, or where $f'(c)$ does not exist.

Critical points

Definition A critical point of a function f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

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EXAMPLE $f(x) = x^3 - 9x^2 + 15x$: $\quad f'(x) = 3x^2 - 18x + 15 = 3(x-1)(x-5)$.

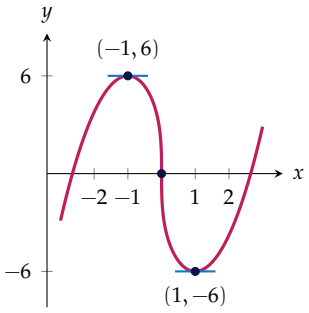
f' exists everywhere, so the critical points are the zeros of f': 1 and 5.

A critical point where f' does not exist

EXAMPLE $g(x) = \sqrt[3]{x}\,(x^2 - 7) = x^{1/3}(x^2 - 7)$, domain $\mathbb{R}$.
For $x \neq 0$, the product rule:

$$g'(x) = \tfrac{1}{3}x^{-2/3}(x^2 - 7) + x^{1/3} \cdot 2x = \frac{(x^2 - 7) + 6x^2}{3x^{2/3}} = \frac{7(x^2 - 1)}{3\sqrt[3]{x^2}},$$

which is 0 exactly when $x = -1$ or $x = 1$.



$y = g(x)$: horizontal tangents at $x = \pm 1$

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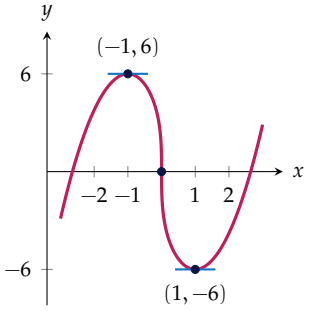
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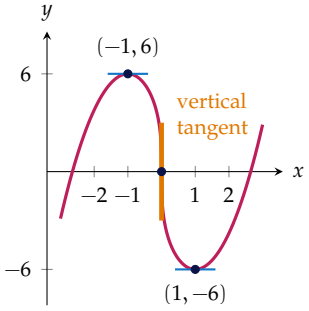
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At 0: a vertical tangent, and no extreme value (g changes sign there). Critical points are candidates, not answers.



$y = g(x)$: horizontal tangents at $x = \pm 1$; a vertical tangent at 0

Question 40

What are the critical points of $f(x) = \sqrt[3]{x^2}\,(x-10)$?

A. 4 only

B. 0 and 10

C. 0 and 4

D. 0 only

In your team

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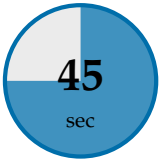
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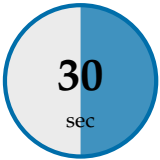
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A B C D

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Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

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Answer

C. For $x \neq 0$: $f'(x) = \frac{2}{3}x^{-1/3}(x-10) + x^{2/3} = \dfrac{2(x-10)+3x}{3\sqrt[3]{x}} = \dfrac{5(x-4)}{3\sqrt[3]{x}}$, zero at 4. At 0: $\lim_{h\to 0}\dfrac{f(h)-f(0)}{h} = \lim_{h\to 0}\dfrac{h-10}{\sqrt[3]{h}}$ does not exist, and 0 is in the domain. A forgets where f' does not exist. B gives the zeros of f, not of f'. D multiplies the derivatives of the two factors: $\frac{2}{3}x^{-1/3}\cdot 1$ is never 0.

The Closed Interval Method

The Closed Interval Method To find the *absolute* maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f at the critical points of f in (a, b).

2. Find the values of f at the endpoints of the interval.

3. The largest of the values from Steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

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Critical points outside $[a, b]$ play no part; and no picture is needed.

The method at work

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1. $f(1) = 7, \quad f(5) = -25.$

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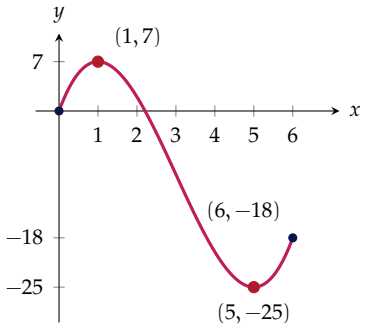
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2. $f(0) = 0$, $f(6) = -18$.

3. Absolute maximum $f(1) = 7$; absolute minimum $f(5) = -25$ — both at critical points; neither endpoint gives an extreme value here.

On frames 3–4 the maximum was at an endpoint: check both kinds of candidate — the critical points and the endpoints.



the picture agrees: $y = x^3 - 9x^2 + 15x$, $0 \le x \le 6$

Question 41

f is continuous on $\mathbb{R}$, its only critical points are 2, 4 and 6, and

$$f(1) = 3, \qquad f(2) = 7, \qquad f(4) = -2, \qquad f(5) = 8, \qquad f(6) = 10.$$

What is the absolute maximum value of f on the interval $[1, 5]$?

A. 8 **B.** 10

C. 7 **D.** 5

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60 sec

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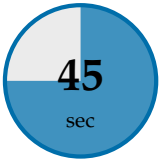
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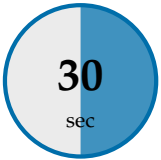
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C. 7 **D.** 5

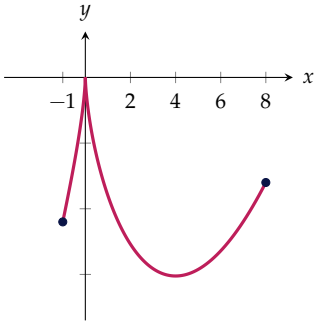
Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

A. The Closed Interval Method on $[1, 5]$: the critical points in $(1, 5)$ are 2 and 4, with values 7 and -2; the endpoints give $f(1) = 3$ and $f(5) = 8$. The largest is $8 = f(5)$. B uses $f(6)$: 6 is a critical point, but not in $[1, 5]$. C stops after Step 1 — the endpoints forgotten. D gives *where* the maximum is, $x = 5$, not its value.

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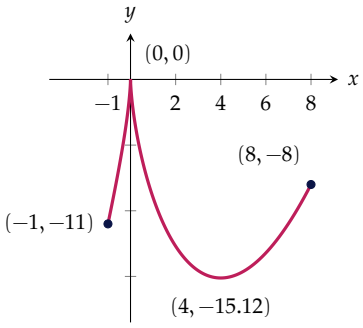


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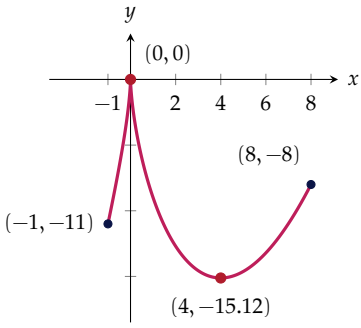
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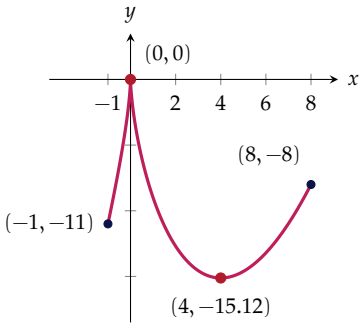
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Solve only $f'(x) = 0$, and the candidates are 4, -1, 8: the "maximum" would be -8 — wrong.



a cusp at the top: $f'(0)$ does not exist

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This week:

Homework: §3.1 — read and work out the examples.

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *"To find the absolute maximum of a continuous function on $[a, b]$ it is not enough to solve $f'(x) = 0$, because …" (one sentence, or a picture).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.